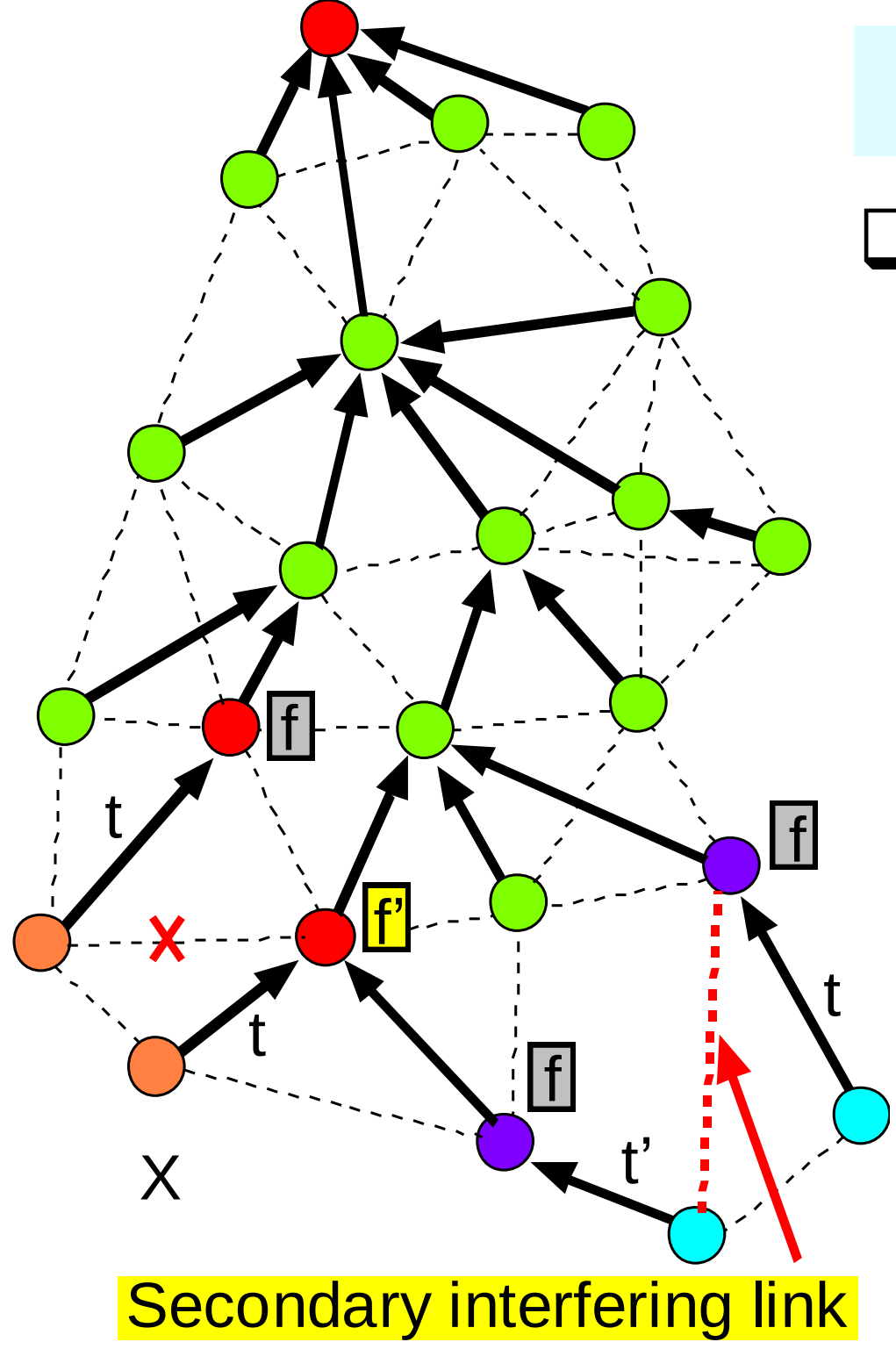


Multi-Channel Scheduling Algorithms for Fast Aggregated Convergecast in Sensor Networks

Amitabha Ghosh, Ozlem Durmaz Incel, V.S. Anil Kumar, and Bhaskar Krishnamachari

Motivation



Maximize Aggregated Sink Throughput

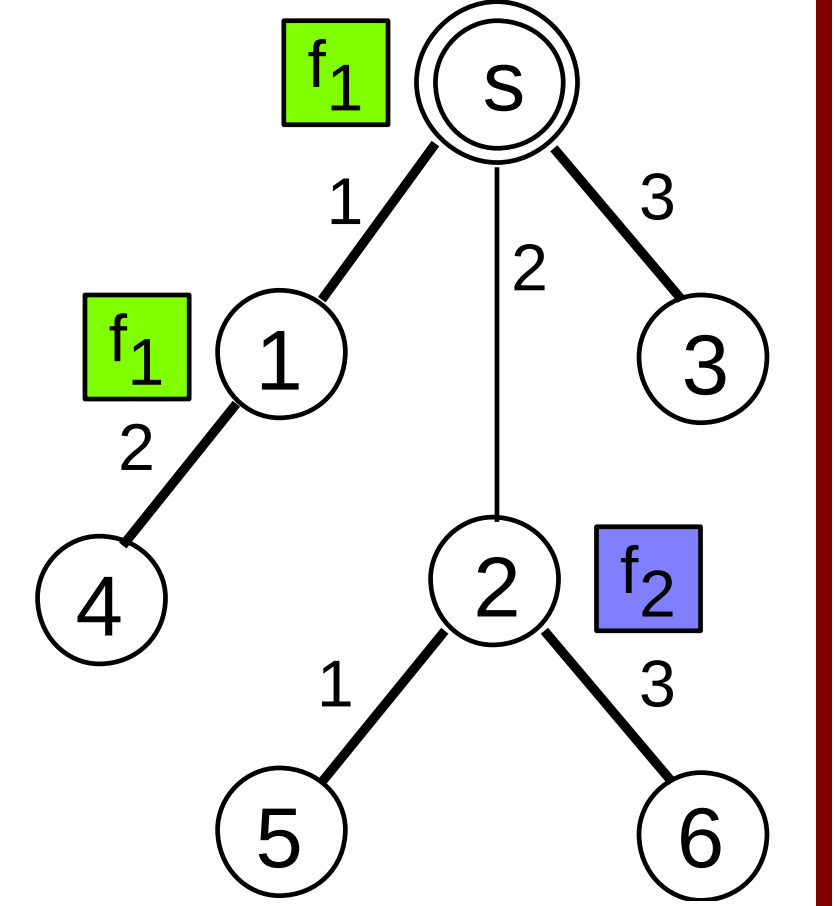
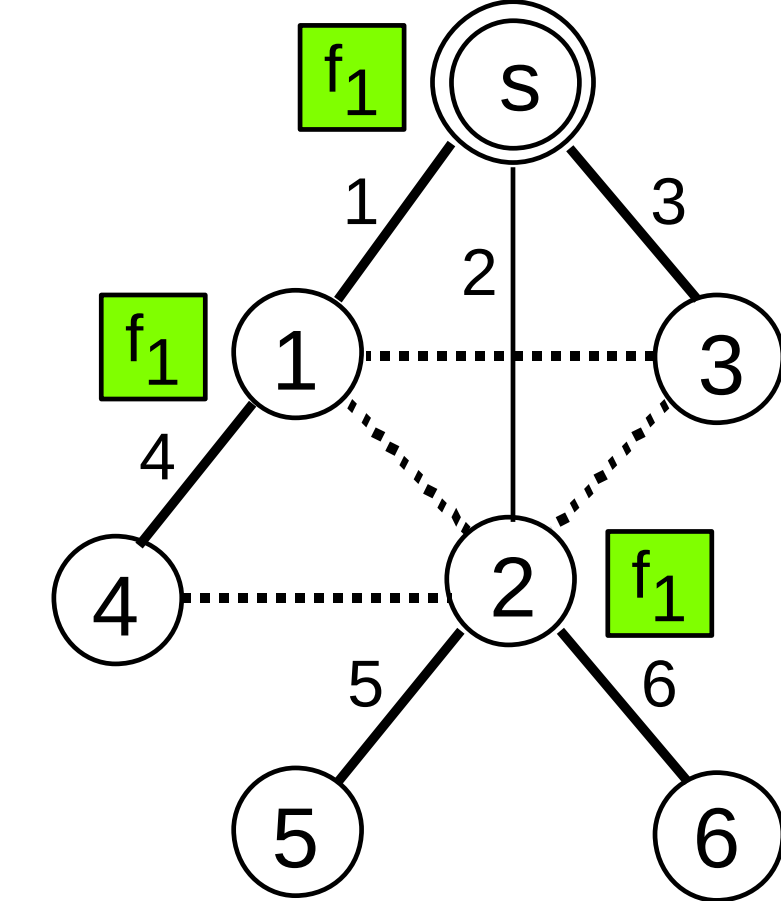
Applications

- High data-rate periodic monitoring (e.g., SHM)
- Real-time, mission-critical, continuous monitoring (e.g., wildfire, security-surveillance, environmental)

Link Scheduling Challenges

TDMA Scheduling

- Half-duplex transceivers (primary conflict)
- Interfering links (secondary conflict)



	Frame 1						Frame 2					
	1	2	3	4	5	6	1	2	3	4	5	6
s	1	2	3	-	-	-	(1,4)	(2,5,6)	3	-	-	-
1	-	-	-	4	-	-	-	-	-	4	-	-
2	-	-	-	-	5	6	-	-	-	-	5	6

	Frame 1			Frame 2		
	1	2	3	1	2	3
s	1	2	3	(1,4)	(2,5,6)	3
1	-	4	-	-	4	-
2	5	-	6	5	-	6

Aggregated Convergecast Schedule Length: 6 time slots with single frequency, and 3 time slots with two frequencies.

Problem Formulation and Solution

Joint Frequency and Time Slot Assignment

Given a routing tree T rooted at the sink on an *arbitrarily* deployed network modeled as a graph $G = (V, E)$, and K orthogonal frequencies:

- Assign a frequency to each of the receivers in T
- Assign a time slot to each of the edges in T

such that the schedule length is minimized.
(NP-hard, Reduction from **Vertex Color**)

Non-uniform Radio Range

Frequency Assignment: A 0-1 ILP Formulation

Goal: Minimize the maximum number of edges that interferes with a given edge.

- Define indicator variables X_{vk} for each edge $e=(u,v)$
 $X_{vk} = 1$, if v is assigned frequency f_k ; 0, otherwise.
- Define $I(e)$ to be the set of edges either adjacent to e , or has secondary conflict with e , and $I_{\geq}(e) \subseteq I(e)$ as the set of edges with larger disks than e .

Minimize λ

s.t.

$$\begin{aligned} \forall e, \forall f_k: \quad & \sum_{v'} |I_{\geq}(e)| X_{vk} \leq \lambda \\ \forall e: \quad & \sum_k X_{vk} = 1 \\ \forall e, \forall f_k: \quad & X_{vk} \in \{0, 1\} \end{aligned}$$

Theorem

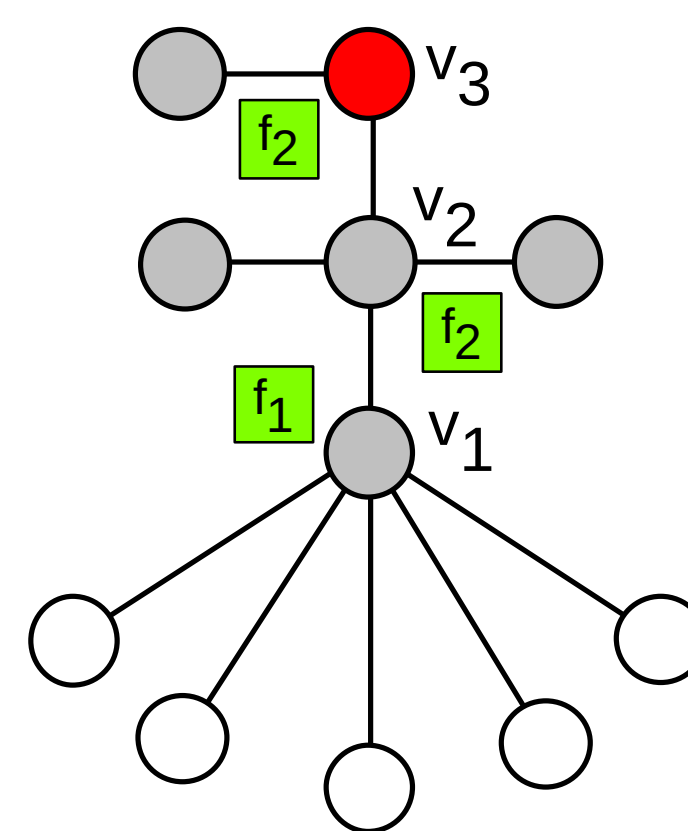
Using a randomized rounding strategy to the linear relaxation of the above ILP, and greedily scheduling longer edges first gives an $O(\Delta(T) \log n)$ approximation to the optimal schedule length on a General Disk Graph.

[1] A. Ghosh, O.D. Incel, V.S. Anil Kumar, and B. Krishnamachari, "Multi-Channel Scheduling Algorithms for Fast Convergecast in Sensor Networks", MASS '09, pp. 363-372.
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Uniform Radio Range: A Two Phase Approach $\mathcal{A}$

Greedy (Load-Balanced) Frequency Assignment

Goal: Assign frequencies so that the maximum number of nodes transmitting on the same frequency (load) is minimized.

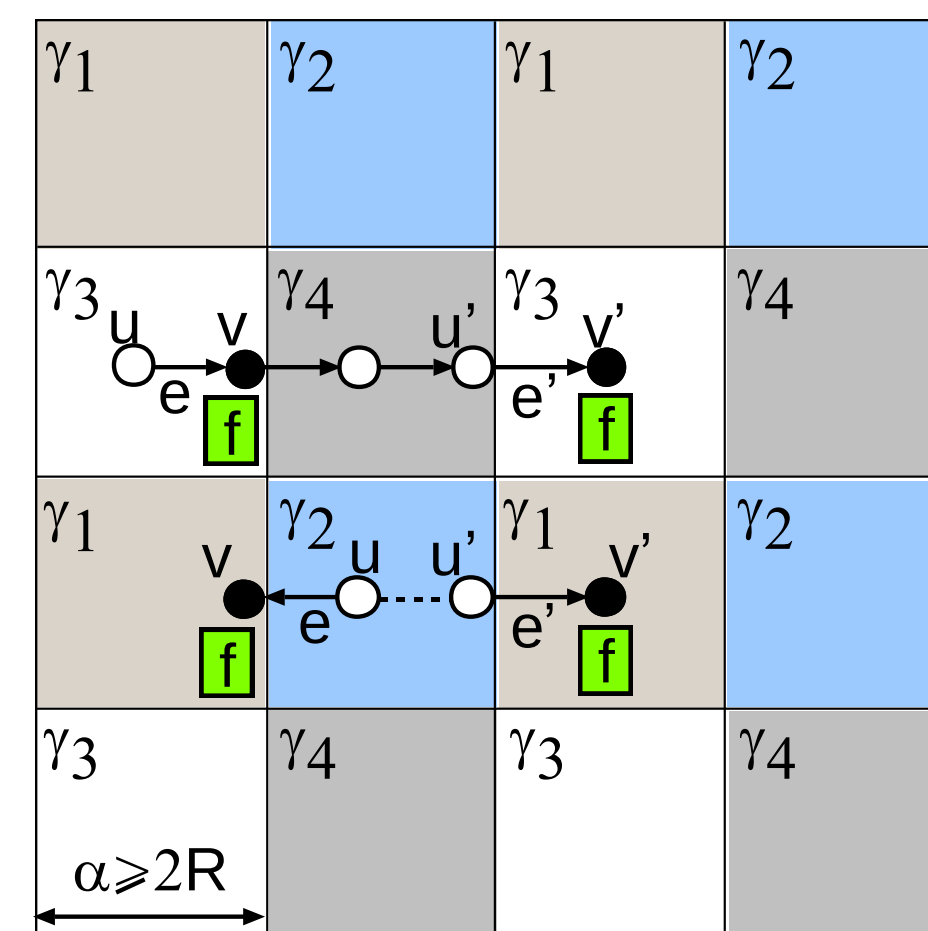


- Order the nodes in decreasing order of their in-degrees: $deg(v_1) \geq \dots \geq deg(v_n)$
- Starting from v_1 , assign successive node a frequency that has the lowest load on it so far

Lemma

Algorithm Frequency-Greedy gives a $(4/3 - 1/3K)$ approximation to the min-max load.

Greedy Time Slot Assignment



Lemma

The minimum schedule length Γ for the whole network is bounded by $4 \max |\gamma_c|$, where γ_c is the set of slots needed to schedule all the links in cell c .

Lemma

If L_c denote the maximal load on any frequency in c , then any greedy time slot assignment can yield $|\gamma_c| \leq 2L_c$

Theorem

Algorithm $\mathcal{A}$ gives a **constant factor $8\mu_\alpha(4/3 - 1/3K)$** approximation to the optimal schedule length on a Unit Disk Graph, where μ_α is the maximum number of edges in any cell that can be scheduled concurrently on the same frequency.

